

Git Command Cheat Sheet

Setup & Configuration

git config --global user.name "Your Name" — Set your Git username (before first commit)
git config --global user.email "you@example.com" — Set your Git email (before first commit)
git config --global color.ui auto — Enable colored output for better readability

Creating & Cloning Repositories

git init — Initialize a new repository (start new project)
git clone <repo_url> — Clone an existing repo (start working on existing project)

Checking Status

git status — Show modified, staged, untracked files
git log — Show commit history
git diff — Show changes not yet staged

Adding & Committing

git add <file> — Stage a specific file
git add . — Stage all changes
git commit -m "message" — Commit staged changes
git commit -am "message" — Stage tracked files + commit in one step

Branching

git branch — List branches
git branch <name> — Create a new branch
git checkout <branch> — Switch to branch
git checkout -b <branch> — Create & switch to a branch
git merge <branch> — Merge another branch into current

Remote Work

git remote -v — Show remote repos
git push origin <branch> — Push commits to remote
git pull — Fetch & merge remote changes
git fetch — Fetch changes only

Undoing Changes

git checkout -- <file> — Revert a file to last commit
git reset HEAD <file> — Unstage a file
git reset --hard — Discard all changes
git revert <commit> — Undo a commit safely

Temporary Work & Stashing

git stash — Save current changes temporarily
git stash apply — Reapply last stash
git stash pop — Reapply and remove stash
git stash list — Show all stashed changes

Undo Commits & History

git reset --soft <commit> — Move HEAD, keep staged changes
git reset --mixed <commit> — Move HEAD, keep unstaged changes
git reset --hard <commit> — Reset everything to a commit
git reflog — Show history of HEAD

Tags & Releases

git tag <name> — Mark a commit (release)
git push origin <tag> — Push tag to remote

Advanced / Cleanup

git clean -f — Remove untracked files
git rm <file> — Remove a tracked file
git mv <old> <new> — Rename file keeping history